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A MINI PROJECT REPORT

on

FinGuard: Hybrid Attention GNN for Fraud Detection

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by

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ABSTRACT

In today's rapidly evolving digital landscape, financial fraud has become more sophisticated and challenging to detect. Fraudsters exploit complex transaction networks to bypass traditional detection systems, which typically analyze transactions in isolation. As a result, these methods fail to capture hidden relationships between entities such as users, merchants, and accounts, allowing complex fraud patterns like mule accounts, circular transactions, and multi-step transfers to go unnoticed. To overcome these limitations, recent research models transaction data as a graph, where entities are represented as nodes and transactions as edges. Graph Neural Networks (GNNs), particularly encoder–decoder architectures, are used to learn relational patterns through message passing, enabling the detection of structural dependencies between entities. Additionally, techniques such as weighted loss functions are employed to address class imbalance, improving the model's ability to detect fraudulent transactions by combining behavioural and relational information. Building on this foundation, we propose a hybrid attention-based heterogeneous Graph Neural Network integrated with an autoencoder for enhanced fraud detection. By extending the graph to a heterogeneous structure, the model captures richer interactions among multiple entity types. The attention mechanism prioritizes important relationships, while the autoencoder identifies anomalies by learning normal transaction behaviour and detecting deviations. This dual-objective framework, combining supervised classification and unsupervised anomaly detection, significantly improves the detection of both known and emerging fraud patterns in complex financial networks.

Keywords: Financial Fraud Detection, Graph Neural Networks, Heterogeneous Graphs, Attention Mechanism, Autoencoder, Encoder-Decoder, Anomaly Detection.

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